



Solar System

Habitability

Defining Habitability

- What do we mean when we say habitable?
 - Earth-like animal life: specific requirements
 - Microbial life - broader set of conditions

Defining Habitability

- What do we mean when we say habitable?
 - Earth-like animal life: specific requirements (oxygen, water, dry land, temperature range)
 - Microbial life - broader set of conditions (more extreme conditions ok)

Common basic requirements for life

- Water, biogenic elements, energy (but elements and energy are common, so liquid water is the key)
- Stable climate, perhaps for long periods

What stabilizes the climate?

- Size - long-term heat source
- Stellar evolution - incoming solar energy
- Impact rate - could result in climate change
- Presence of large, natural satellite - prevents large swings in obliquity
- Oceans - regulate global temperatures

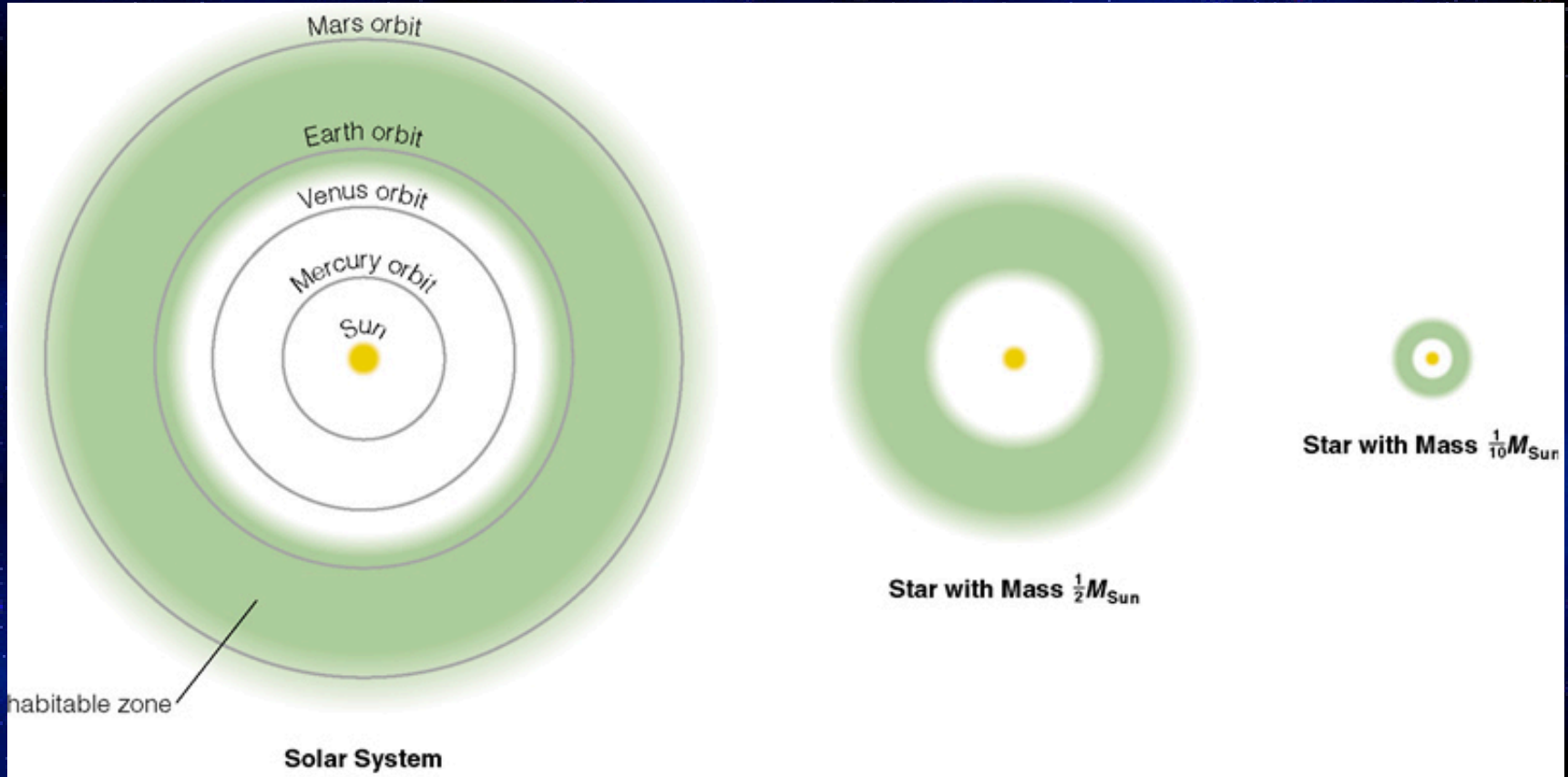
What does life need?

- Elements – Life on Earth uses about 25 of the 92 naturally occurring elements.
- Energy – An energy source for metabolism (from sun, chemical reaction, etc.)
- Liquid Water – dissolves organic molecules, transports chemicals, involved in metabolic reactions.
 - Could you use something other than water? Maybe. Water is good because:
 - It stays liquid over a large range of temperatures.
 - It expands (floats) as it freezes.
 - Electrical charge distribution makes it polar. (affects what dissolves in water, cell membranes don't)

Solar System Habitability

- Moon and Mercury – least likely
- Venus – highly unlikely
- Jovian Planets – Unlikely
- Small moons, asteroids, comets, Pluto – Unlikely
- Mars and large moons – most likely
 - These are places where what we think life needs may be in place.

A Habitable Zone



A region of our Solar System where temperatures (in principle) would allow for liquid water to exist on the SURFACE of a planet (of the correct type).

Comparative Planetology: Venus, Earth, and Mars

The Greenhouse Effect on the Terrestrial Worlds

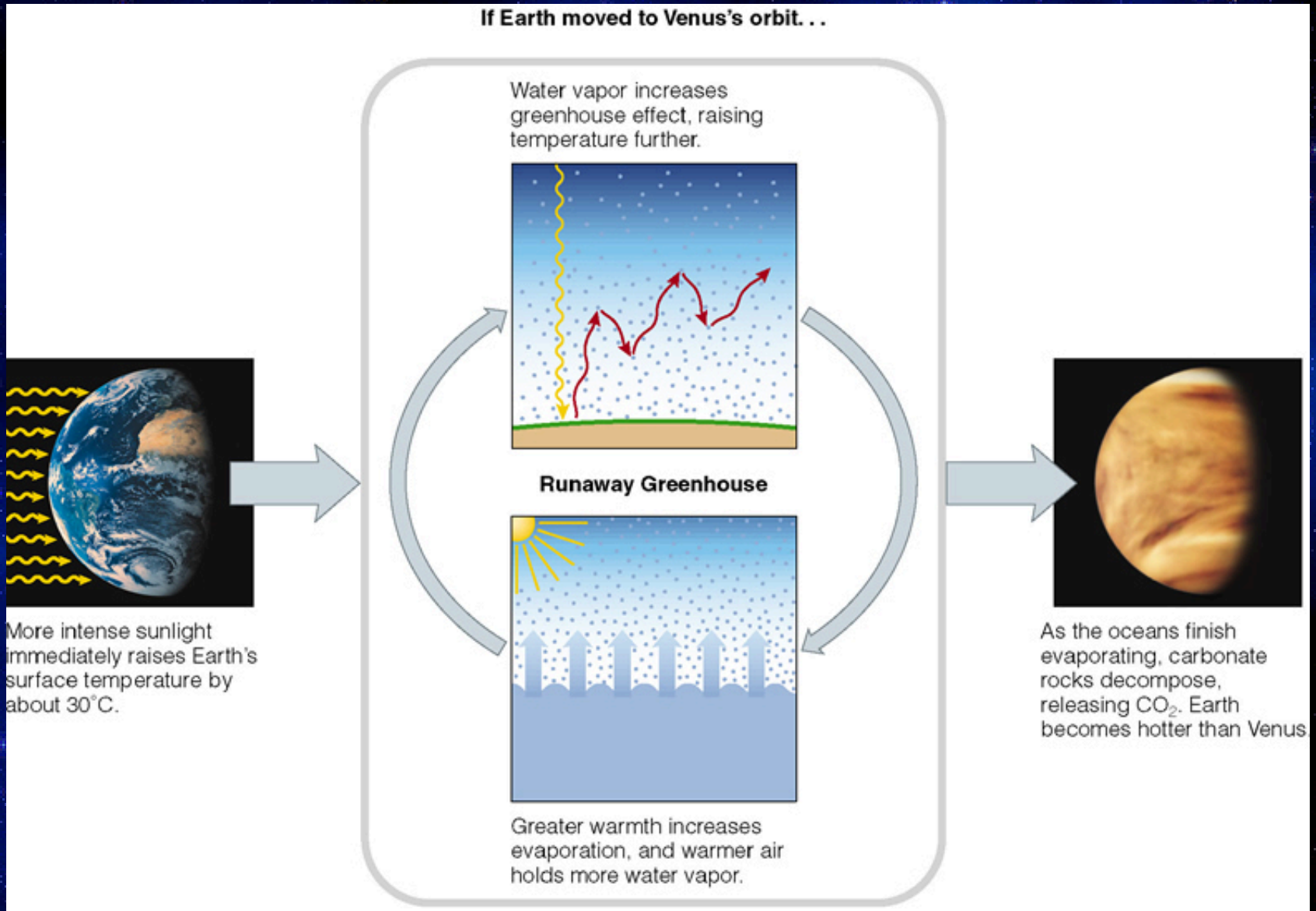
World	Actual Average Surface Temperature	"No Greenhouse" Average Surface Temperature*	Greenhouse Warming (actual temperature minus "no greenhouse" temperature)
Venus	470°C	−43°C	513°C
Earth	15°C	−17°C	32°C
Mars	−50°C	−55°C	5°C

The Greenhouse Effect makes a large difference in the Venusian atmosphere, but very little difference in the Martian atmosphere.

Venusian Oceans?

- No water today! (no ice or water, almost no vapor)
- Either:
 - Venus didn't have much water (unlikely), or
 - Venus lost it. (how?)
- Venus likely had lots of water (or vapor) in its youth.
 - No ozone layer, UV would break water
 - Check on amount of Deuterium!

Runaway Greenhouse Effect



Surface Habitability Factors

- Distance to the Sun (Venus got too warm and had a runaway greenhouse)
- Planetary Size (Mars cooled too quickly, geologically inactive)
- Atmospheric Loss (Mars has no magnetic field protection and low level of volcanism for replacement, Venus is losing too)

Outside the Zone

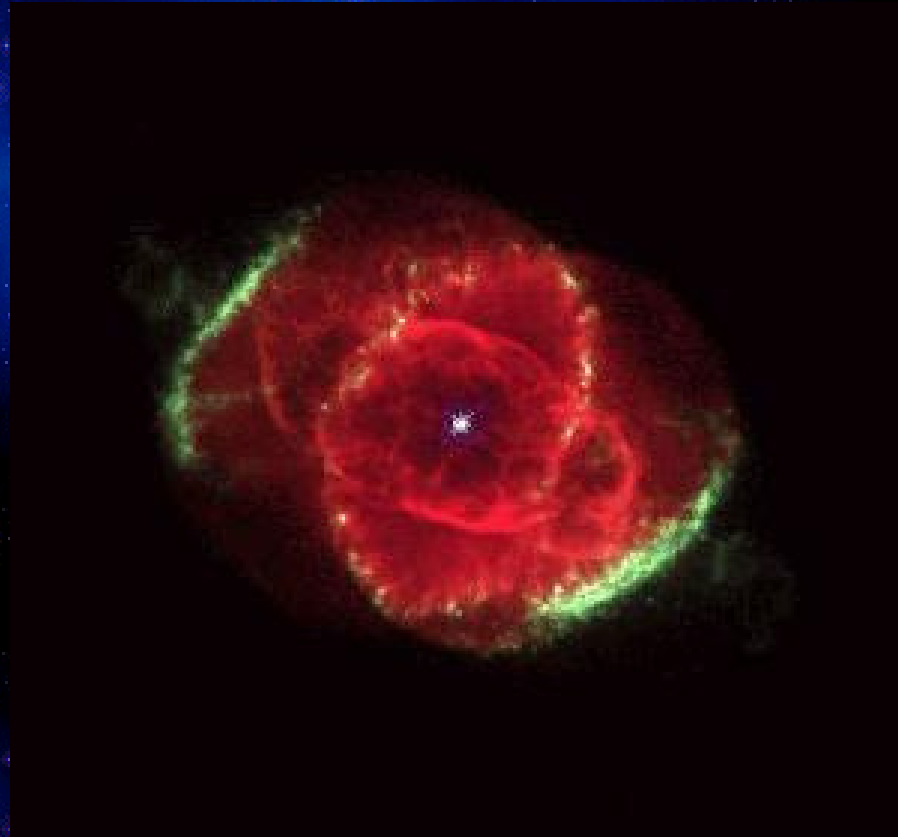
- The concept of a habitable zone
 - precedes discussions of life on Europa or Titan
 - Only took into account liquid water AT THE SURFACE
 - Only talked about heating by the Sun
 - Only used the liquid water

Future Habitability

- Habitability WILL end on Earth.
- As the Sun brightens, we'll lose water.
- In 3-4 billion years, Earth will be hot enough for a runaway greenhouse.
- When the Sun dies (about 5 billion years from now), it will become a red giant. ($>100\times$ its present radius) We will bake!
- Then it will eject its outer layers in a minor (!!) explosion which will likely destroy the planet.

Could we survive?

- Only if we move first.
- And if our new planet dies? Keep moving!
- A new star! (beyond our reach at present)
- When it dies, keep moving.
- Maybe we'll make it that long.



THANK YOU!